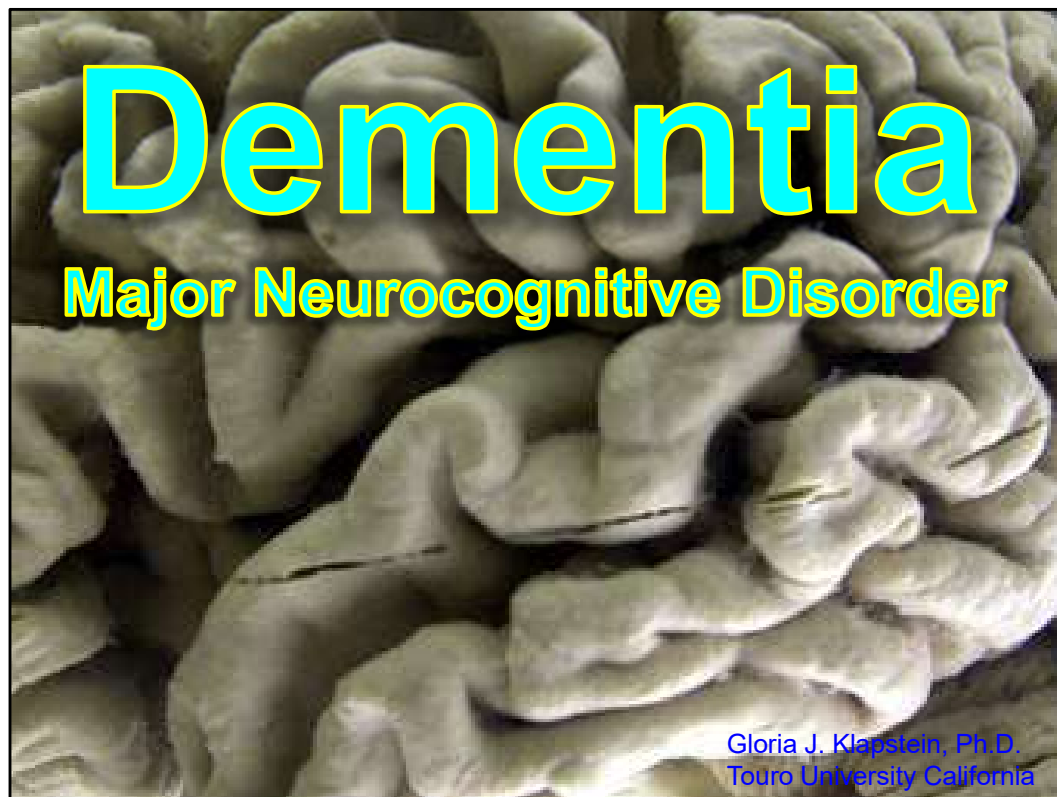






Dementia

- Loss of cognitive function affecting daily life
 - Memory
 - Language
 - Reasoning/Problem Solving } Executive functions
 - Sensory changes
 - Personality/Emotional changes
 - Self care/Management
- DSM-5 calls this **Major Neurocognitive Disorder**
- Progressive course of disease
 - Early/Milder symptoms sometimes referred to as:
 - Mild Neurocognitive Disorder
 - Minor Neurocognitive Disorder
 - Mild Cognitive Impairment (MCI)
- Not a normal part of aging

Dementia Epidemiology

- According to WHO, #people living with dementia:
 - 47.47 million in 2015
 - 75.63 million in 2030
 - 135.46 million in 2050
 - Cost to USA = \$290 billion in 2019
 - After 65 years of age, the lifetime risk of developing dementia is ~20%
 - AD affects 5.8 million Americans and is the 6th leading cause of death. Median survival time after diagnosis is 4.5 years.
-  Risk factors for dementia:
- age
 - family history
 - apolipoprotein E e4 genotype
 - cardiovascular comorbidities
 - chronic anticholinergic use
 - lower educational level

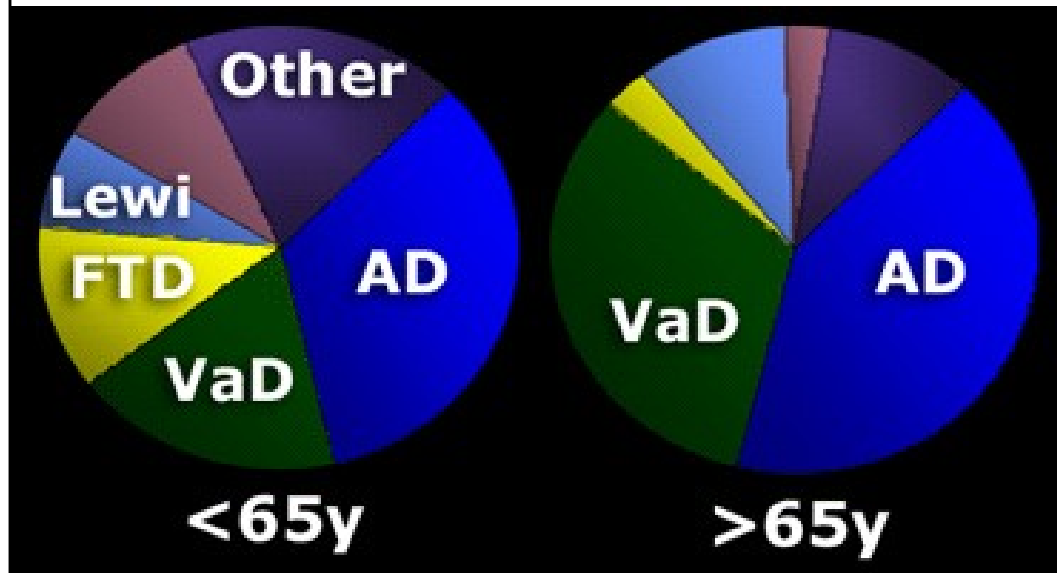
Projected increase
mainly due to
population aging

https://www.who.int/mental_health/neurology/dementia/dementia_thematicbrief_epidemiology.pdf

Simmons, Evaluation of suspected dementia, JAFP, 2011

DSM5: at age 65, 1-2% prevalence of NCD, by age 85, 30%!

Prevalence of Etiology is age-dependent



<http://www.radiologyassistant.nl/en/43dbf6d16f98d>

AD: Alzheimer's disease

VaD: Vascular dementia

FTD: Frontotemporal dementia



Alzheimer's disease

- Age of onset itself has a heritability of around 40%
- most frequent type of dementia in the elderly; affects almost half of all patients with dementia
- 2 - 3% of people at 65 y.o.;
- 25 - 50% of people > 85 y.o..

AD Genetics and A β

Four genes have been conclusively shown to affect the development of Alzheimer's disease: the APP gene on chromosome 21, the PS1 gene on chromosome 14, the PS2 gene on chromosome 1, and the apoE gene on chromosome 19. Mutations of APP, PS1 and PS2 genes are linked to the rare early-onset form of familial AD. Many people with Down's syndrome also develop AD-like dementia by the age of 40. They have three copies of chromosome 21 (where the APP gene is located), instead of two for normal people.

The APP gene encodes the precursor protein to beta amyloid. One more copy of the APP gene in people with Down's syndrome should make more precursor proteins, thereby producing more A β peptides. Mutations of the APP gene in the early-onset AD either increase the total level of A β peptides or favor the production of A β 42. Both cases increase the chance of forming amyloid plaques.

PS1 and PS2 genes encode presenilin 1 and presenilin 2, respectively. They are the components of the γ -secretase, which cleaves APP to generate A β . Their mutations favor the production of A β 42.

The risk gene with the strongest influence is called apolipoprotein E-e4 (APOE-e4). Scientists estimate that APOE-e4 may be a factor in 20 to 25 percent of Alzheimer's cases. The apoE gene encodes apolipoprotein E (apoE) which helps carry cholesterol in

the blood.

In April of 2011, 5 new genes were discovered, mutations of which each raise Alzheimer's risk by 15% or less. One of these is CD33, a gene tied to the brain's innate immune system. In some cases, it might not be eliminating as much beta amyloid as it should. In others, it may be too active and trigger inflammation.

Diagnosing Alzheimer's disease



- Clinical diagnosis. Clinical observation, rule out alternative diagnoses.
- Definitive diagnosis: postmortem (histology)
- Biomarkers: some available

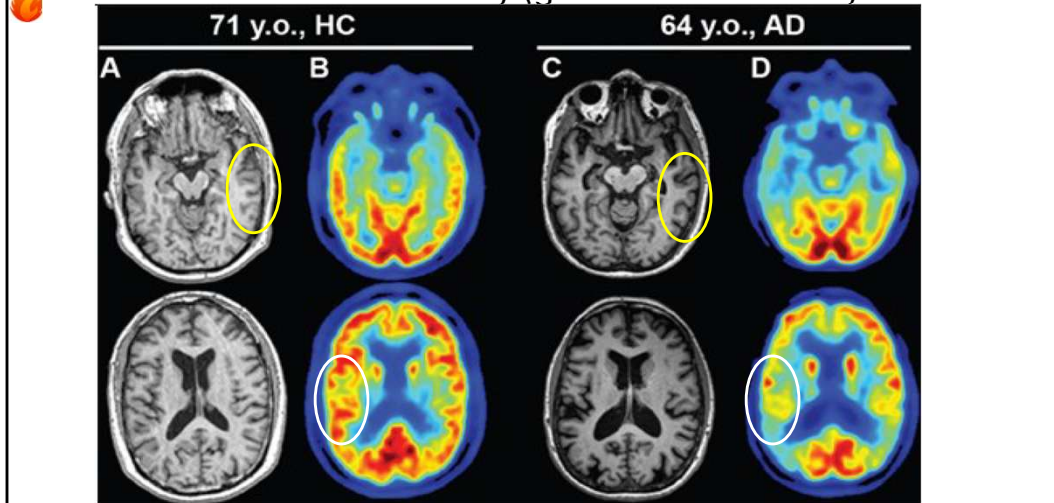
Diagnosing Alzheimer's Disease

- Structural imaging (MRI, CT)

- 🔥 – Brain shrinkage (esp. hippocampus, cortex)

- Functional imaging (PET, fMRI)

- 🔥 – Reduced brain cell activity (glucose metabolism)



Alzheimer's disease (AD). Axial T1-weighted magnetic resonance images of a healthy 71-year-old (A) and a 64-year-old with AD (C). Note the reduction in medial temporal lobe volume in the patient with AD. Fluorodeoxyglucose positron emission tomography scans of the same individuals (B and D) demonstrate reduced glucose metabolism in the posterior temporoparietal regions bilaterally in AD, a typical finding in this condition. HC, healthy control. (Images courtesy of Gil Rabinovici, University of California, San Francisco and William Jagust, University of California, Berkeley.)

FIGURE 25-1, Harrison's Principles of Internal Medicine, 20e, Chapter 25: Dementia
William W. Seeley; Bruce L. Miller

Image credit: University of Pittsburgh Medical Center

http://www.alz.org/research/science/earlier_alzheimers_diagnosis.asp#Brain

Diagnosing Alzheimer's Disease

- Molecular imaging (beta-amyloid radiotracer)



- Pittsburgh compound B (PIB) (images amyloid)

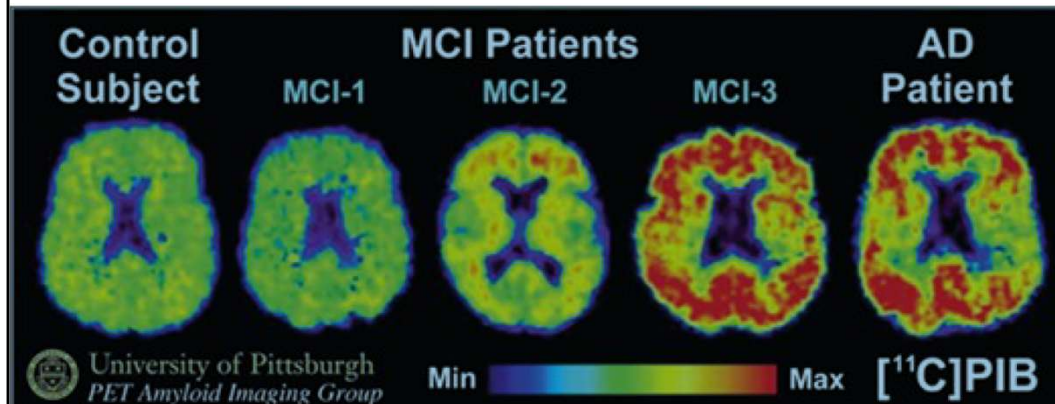
- 18F flutemetamol (flute)

- Florbetapir F 18 (18F-AV-45)

- Florbetaben (BAY 94-9172)



- Tauvid (flortaucipir F18) approved May 28, 2020 for presymptomatic use



MCI – Mild cognitive impairment

PIB images amyloid

Tauvid images tau

Diagnosing Alzheimer's Disease

- CSF/Urine/Blood Tests



Lumipulse – approved May 2025

- Measures ratio of pTau217 and amyloid beta(1-42) in blood plasma
- Can positively test for or rule out AD
- Negative test correctly rules out AD 97% of the time
- Positive test correctly predicts AD 92% of the time

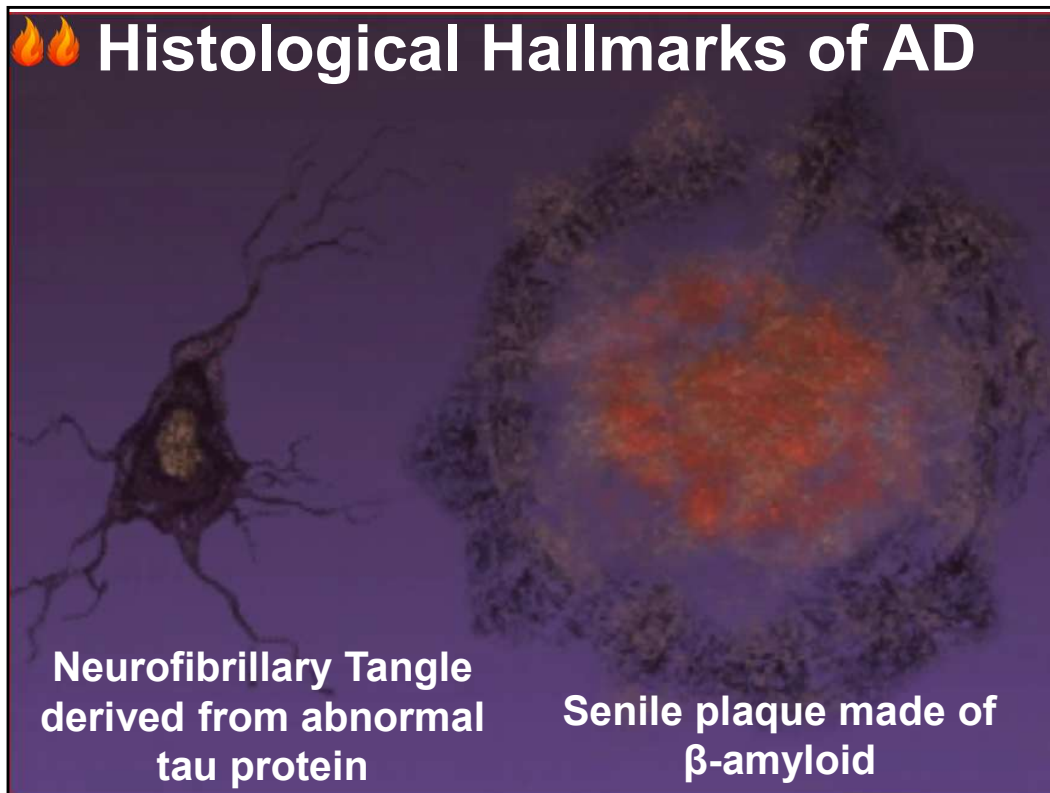


Elecsys pTau181 – approved October 2025

- Measures phosphorylated tau in blood plasma
- Used to rule out AD if patient has cognitive decline
- Negative test correctly rules out AD 98% of the time

<https://www.fda.gov/news-events/press-announcements/fda-clears-first-blood-test-used-diagnosing-alzheimers-disease>

<https://www.nature.com/articles/d41586-025-03394-w>



Ab is a highly hydrophobic peptide, which aggregates to form oligomers. If these oligomers aggregate further, they start forming fibers, which eventually precipitate and accumulate in the disease-defining amyloid plaques. While the fibers are deposited in the brain parenchyma, additional toxic aggregation events appear to be induced within the cell. Tau, a protein that stabilizes microtubules, dissociates from the cytoskeleton and aggregates within the soma of neurons to form the intracellular tangles (Hardy and Selkoe, 2002). This process may then initiate a variety of less understood toxic insults, ultimately leading to the tremendous neuronal loss observed in AD brains. This process, which appears to be initiated by the oligomerization of Ab, is now known as the amyloid cascade (Hardy and Selkoe, 2002).



Neurofibrillary Tangle



Senile plaque

<http://www.nealhendrickson.com/mcdougall/2004nl/040600pualzheimer.htm>

Alzheimer's disease

• Genetic Factors - Late Onset

-  **APOE e2.** This is the least common form of the gene. It reduces the risk of AD
-  **APOE e4.** More common form. Increases the risk of AD, more-severe AD.
-  **APOE e3.** Most common form. It doesn't seem to affect the risk of AD.
 - Apolipoproteins bind and transport lipids; cholesterol, fat-soluble vitamins, etc. in blood/CSF/lymph
- **ABCA7.** Some forms increase risk of AD. Role in how the brain uses cholesterol.
- **CLU.** Helps brain clear out beta-amyloid.
- **CR1.** Involved in immune system function. Low levels increase inflammation.
- **PICALM.** Involved in neuronal communication and beta-amyloid clearance.
- **PLD3.** Physiologic role unknown; linked to a significantly increased risk of AD.
- **TREM2.** Role in inflammatory response. Rare variants cause increased AD risk.
- **SORL1.** Role in beta-amyloid production. Some forms of SORL1 on chromosome 11 linked to AD.

<https://en.wikipedia.org/wiki/Apolipoprotein>

Apolipoprotein E (apoE) plays an important role in the transport and uptake of [cholesterol](#) by way of its high affinity interaction with lipoprotein receptors, including the [low-density lipoprotein](#) (LDL) [receptor](#). ApoE is the major lipoprotein in the [central nervous system](#). Recent findings with apoA1 and apoE suggest that the [tertiary structures](#) of these two members of the [human](#) exchangeable apolipoprotein [gene family](#) are related.^[8] The three-dimensional [structure](#) of the LDL receptor-binding [domain](#) of apoE indicates that the protein forms an unusually elongated four-helix bundle that may be stabilised by a tightly packed [hydrophobic core](#) that includes [leucine zipper](#)-type interactions and by numerous [salt bridges](#) on the mostly charged surface. [Basic amino acids](#) important for LDL [receptor binding](#) are clustered into a surface patch on one long [helix](#).^[9]

Alzheimer's disease

- **Genetic Factors - Early Onset** (30-60 years old)



Amyloid precursor Protein (APP).

Presenilin 1 (PSEN1)

Presenilin 2 (PSEN2)

- Presenilins are catalytic components of secretase protease complexes which cleave amyloid precursor protein

AD Genetics and A β

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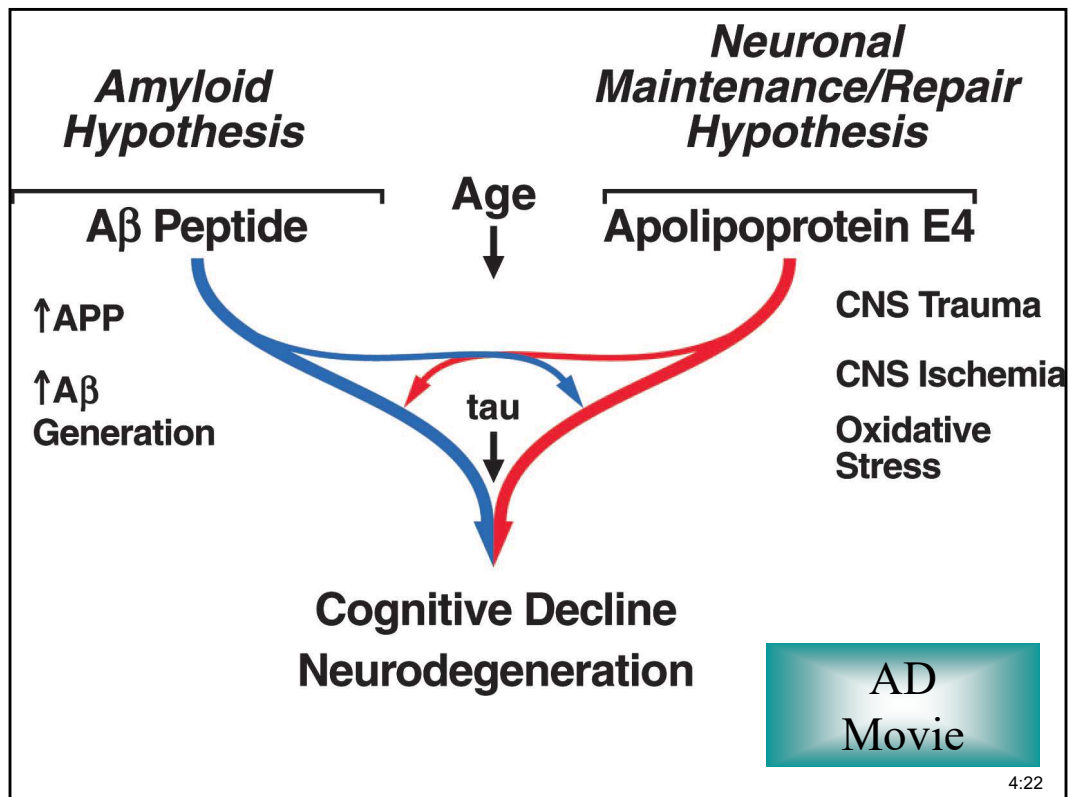
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The apoE gene encodes apolipoprotein E (apoE) which helps carry cholesterol in the blood.

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Alzheimer's risk by 15% or less. One of these is CD33, a gene tied to the brain's innate immune system. In some cases, it might not be eliminating as much beta amyloid as it should. In others, it may be too active and trigger inflammation.

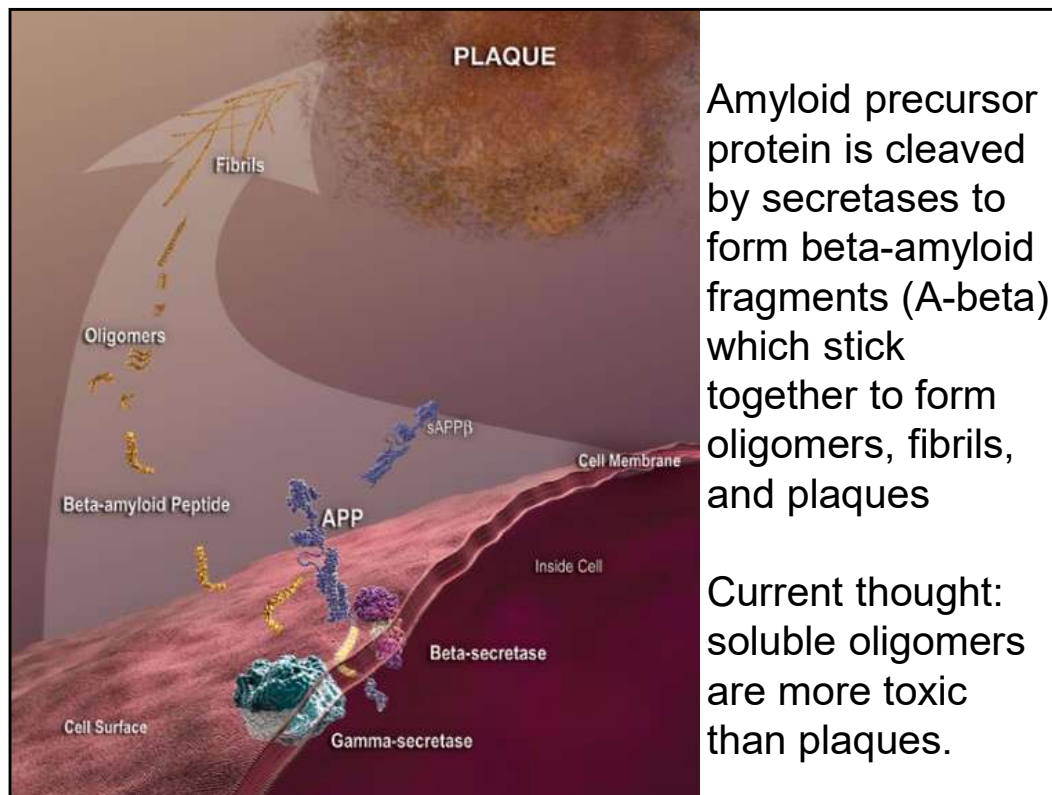
Using a relatively new approach called **genome-wide association study (GWAS)**, researchers have identified a number of genes in addition to APOE ε4 that may increase a person's risk for late-onset Alzheimer's, including BIN1, CLU, PICALM, and CR1.



Mahley2006PNAS

Fig. 1. Multiple factors interact through various pathways to cause cognitive decline and neurodegeneration. ApoE4 may be associated with neuropathology through interactions with A-beta peptide that converge on the amyloid cascade or may act independently of A-beta.

ApoE has critical functions in redistributing lipids among CNS cells for normal lipid homeostasis, repairing injured neurons, maintaining synaptodendritic connections, and scavenging toxins. In multiple pathways affecting neuropathology, including Alzheimer's disease, apoE acts directly or in concert with age, head injury, oxidative stress, ischemia, inflammation, and excess amyloid peptide production to cause neurological disorders, accelerating progression, altering prognosis, or lowering age of onset. We envision that unique structural features of apoE4 are responsible for apoE4-associated neuropathology. Although the structures of apoE2, apoE3, and apoE4 are in dynamic equilibrium, apoE4, which is detrimental in a variety of neurological disorders, is more likely to assume a pathological conformation. ... In response to CNS stress or injury, neurons can synthesize apoE. ApoE4 uniquely undergoes neuron-specific proteolysis, resulting in bioactive toxic fragments that enter the cytosol, alter the cytoskeleton, disrupt mitochondrial energy balance, and cause cell death.



The beta amyloid is a protein fragment snipped from a larger protein called amyloid precursor protein (APP). These fragments clump together and are mixed with other molecules, neurons, and non-nerve cells. In AD, plaques develop in the hippocampus, a structure deep in the brain that helps to encode memories, and in other areas of the cerebral cortex that are used in thinking and making decisions.

Beta amyloid plays a central role in the development of AD. Its insights let us understand the effects of [cholesterol and apolipoprotein E](#) on AD. The early-onset familial AD that occurs between the ages of 30 and 60 has also been demonstrated to result from mutations of the genes involved in the production of beta amyloid. For this reason, beta amyloid is an important target in the development of new drugs against AD.

From APP to Beta Amyloid (Aβ)

The amyloid precursor protein (APP) appears to be important in helping neurons grow and survive. APP may help damaged neurons repair themselves and may help parts of neurons grow after brain injury. After it is made, APP sticks through the neuron's membrane, partly inside and partly outside the cell.

The beta amyloid (Aβ) is generated by two enzymes which cut APP at two different positions: β-secretase cuts APP at a position outside the cell and γ-secretase cuts APP at a position inside the cell membrane. The resulting Aβ peptide contains about 40 amino acids.

The γ-secretase lacks sequence specificity. Its cleavage site may shift slightly, depending on the molecular structure of the enzyme and its environment. As a result, the length of Aβ ranges from 39 to 43 amino acids. It has been found that the longer one is more likely to form plaques than the shorter one. Researchers usually use two Aβ peptides to compare their toxicities: Aβ with 40 amino acids (represented as Aβ40) and Aβ with 42 amino acids (represented as Aβ42). Aβ42 is more toxic than Aβ40.

The Toxicity of Aβ

Individual Aβ peptides can be cleared away by degradative enzymes such as insulin-degrading enzyme, neprilysin, and plasmin. After they aggregate to form plaques, they are difficult to be removed. The plaques may induce inflammatory responses, creating oxygen free radicals. In addition, they may promote the formation of neurofibrillary tangles, resulting in cell death.

In a normal brain, Aβ40 is produced at higher level than Aβ42. However, the amyloid plaque in Alzheimer's disease is composed primarily of Aβ42. Experiments have also shown that Aβ42 aggregates more rapidly than Aβ40, consistent with other observations that Aβ42 is more toxic than Aβ40.

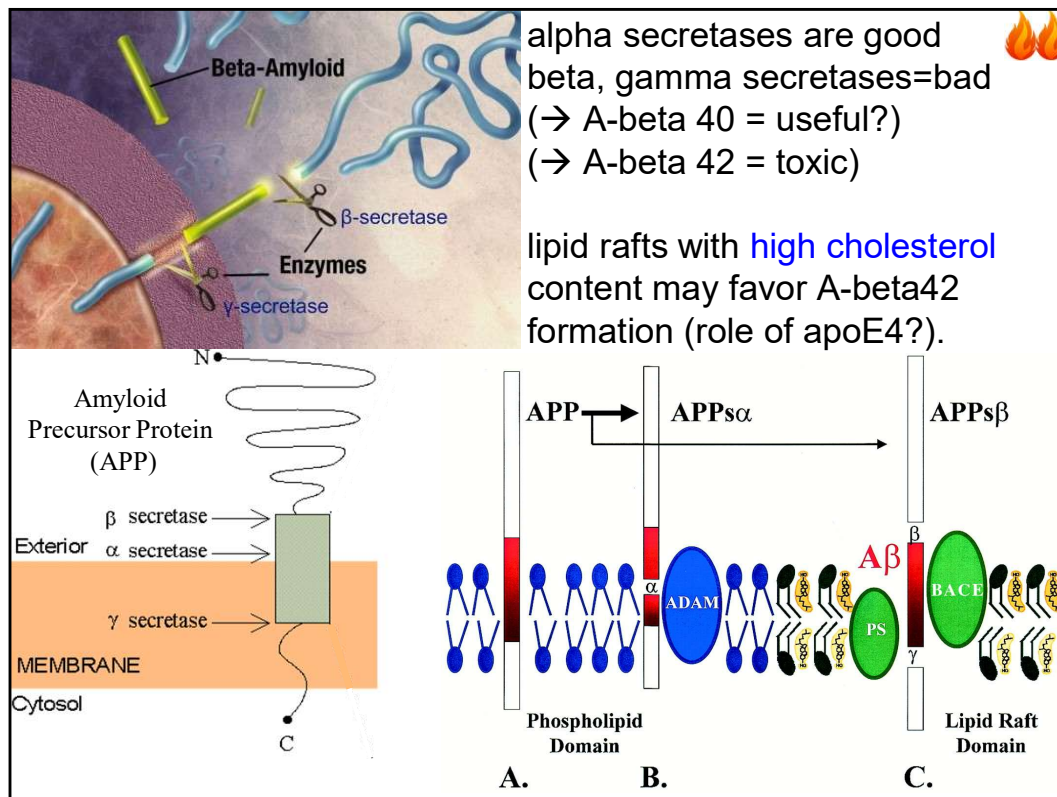
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APP to generate A β . Their mutations favor the production of A β ₄₂.
The apoE gene encodes apolipoprotein E (apoE) which helps carry cholesterol in the blood.



ADAM = A Disintegrin And Metalloproteinase; family of alpha secretases;

BACE = β-site amyloid precursor protein cleaving enzyme

Figure 10-E-1. Generation of the beta amyloid (Aβ) from its precursor protein APP.

The APP gene is located on chromosome 21. It is expressed by alternative splicing, generating three different isoforms, with 695, 751, and 770 residues, respectively. The sequence numbering in this figure refers to the 770 isoform.

APP is a transmembrane protein, with a long extracellular region and a short intracellular region. It contains three proteolytic cleavage sites for α-, β- and γ-secretase, respectively. The sequential cleavages by β- and γ-secretases will generate the **beta amyloid** (Aβ). If APP is cut by α-secretase, no Aβ will be generated, since its cleavage site is located in the middle of the Aβ region.

Mutations of APP around the positions 670, 693 and 715 have been found to increase the risk of Alzheimer's disease. The position 670 is near the β-secretase cleavage site (Aβ starts from D672). A double mutation at K670 and M671 increases the production of both Aβ40 (with 40 residues) and Aβ42 (with 42 residues). Mutations near the γ-secretase cleavage site favors the production of Aβ42. The effects of the mutations around E693 is more complicated (review).

The γ-secretase cleavage site is located in the membrane, which is unusual and more difficult. It is suggested (reference) that the substrate may need to associate with sphingolipids and cholesterol to increase its stability. Recently, it was found that the Aβ protein indeed contains a sphingolipid binding domain (reference). This model can also explain why lowering cholesterol level can reduce the production of Aβ proteins.

Review Article: Take five: BACE and the γ-secretase quartet conduct Alzheimer's amyloid β-peptide generation - EMBO J., 2004.

<http://www.pnas.org/content/98/10/5371/F1.expansion>

Figure 2. A putative model of the processing of APP in relation to the lipid composition of membranes. (A) APP is a transmembrane protein. (B) Cleavage of APP by α secretases, such as ADAM10, produces APPsα, which requires a membrane domain that is cholesterol poor, such as phospholipid domains. (C) Cleavage of APP by the β and γ secretases, BACE, and Presenilin (PS) produces Aβ and APPsβ. This step requires a membrane domain that is cholesterol rich, such as a lipid raft. APP processing can be directed toward α-secretase or β/γ-secretase pathways by modulating the cholesterol content of the membranes. Lipid identities: blue, phospholipid; green, sphingolipids; yellow, cholesterol.

<http://www.web-books.com/eLibrary/ON/B0/B3/04MApoE.html>

High Cholesterol Level Increases Aβ Production

Dozens of reports have confirmed that cholesterol-lowering drugs such as statins can decrease the risk of Alzheimer's disease (AD). It was further shown that low cholesterol reduces the production of beta amyloid (Aβ). The underlying mechanism has been elucidated.

The beta amyloid is generated from its precursor protein (APP) by two enzymatic cleavages: β-secretase cuts APP at a position outside the cell and γ-secretase cuts APP at a position inside the cell membrane (more info). The γ-secretase site is unusual because it is located in the membrane. The intramembrane catalysis requires a domain structure called "lipid raft" which is rich in

cholesterol (figure). Therefore, higher cholesterol level favors the formation of lipid rafts, thereby increasing the production of beta amyloid.

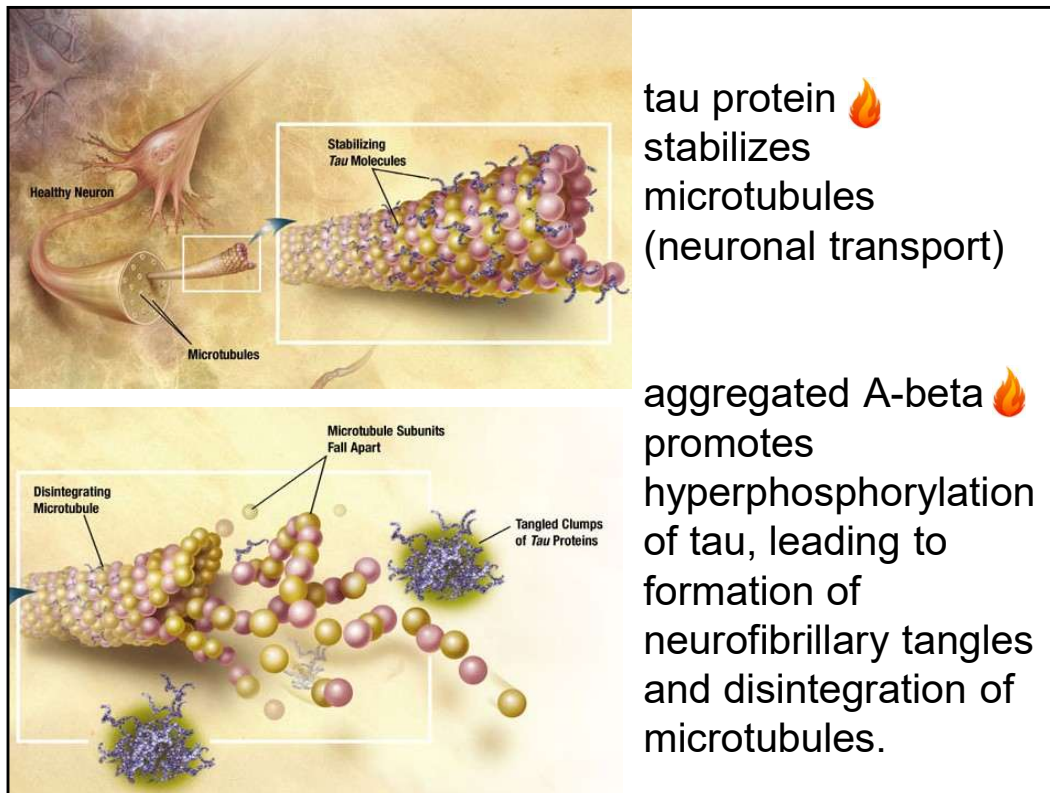
The effects of ApoE on A β

Apolipoprotein E (apoE) is a protein with 299 amino acids. Its gene is located on chromosome 19. The gene has three alleles (variants), encoding three protein isoforms: apoE2, apoE3, and apoE4. Their frequencies in populations are 5-10%, 60-70%, and 15-20%, respectively (reference). People with apoE4 are more likely to develop Alzheimer's disease.

ApoE is important for removing excess cholesterol from the blood, and does so by carrying cholesterol to receptors on the surface of liver cells. Defects in apoE may cause the inability to bind to the receptors, which leads to an increase in a person's blood cholesterol and consequently their risk of atherosclerosis.

In the central nervous system (CNS), ApoE is produced by astrocytes, activated microglia, and neurons. The role of apoE is to redistribute lipid (including cholesterol) among CNS cells. Since cholesterol is involved in the development of AD, the finding that apoE4 is a risk factor for AD might result from its function as a cholesterol transport protein. However, experiments do not support this idea.

ApoE has been shown to affect beta amyloid in several ways. It can bind to A β , thereby inhibiting its clearance and stimulating its deposition. The apoE4 binds to A β more rapidly than apoE3. ApoE can also increase A β production, but apoE4 stimulates A β production more effectively than apoE3.



<http://www.web-books.com/eLibrary/ON/B0/B3/05MTau.html>

The tau protein belongs to a class of proteins known as microtubule-associated proteins (MAPs) which are essential for the stability of **microtubules**. The general function of microtubules in most cells is to transport various nutrients and molecules within the cell. In neurons, it could also play a critical role in memory formation and storage ([more info](#)). In Alzheimer's disease (AD), tau is changed chemically. It begins to pair with other threads of tau and they become tangled up together. When this happens, the microtubules disintegrate, collapsing the neuron's transport system. It could also directly affect memory formation and storage. The early stage of neurofibrillary tangles is associated with the loss of dendritic microtubules and synapses. In later stages, the cell body is also affected, eventually resulting in cell death ([reference](#)).

Excessive phosphorylation of the tau protein has been shown to be the major chemical change that causes its abnormality in AD ([review](#)). Phosphorylation is a process that adds a phosphate group to an amino acid in a protein. The more amino acids are phosphorylated, the more likely a protein's structure and function will be significantly altered. An enzyme called glycogen synthase kinase 3 (GSK3) plays an important role in tau phosphorylation. Aggregated beta amyloid can activate GSK3 to phosphorylate the tau protein ([reference](#)). When both tau protein and beta amyloid were first found to be involved in AD, their cause-effect relationship was not clear. It is now confirmed that beta amyloid causes the abnormality of the tau protein, not the other way around.

Tauopathies

In studying tau and what can go wrong, investigators have found that tau abnormalities are also central to other rare neurodegenerative diseases. These diseases, called tauopathies, include frontotemporal dementia, Pick's disease, supranuclear palsy, and corticobasal degeneration. They share a number of characteristics, but also each have distinct features that set them apart from each other and from Alzheimer's disease. Characteristic signs and symptoms include changes in personality, social behavior, and language ability; difficulties in thinking and making decisions; poor coordination and balance; psychiatric symptoms; and dementia. Recent advances include the discovery of mutations in the tau gene that cause one tauopathy called frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17). The development of several mouse models that produce tau tangles, will allow researchers to address the many questions that remain about these diseases.

References:

Role of Tau Protein in Both Physiological and Pathological Conditions - Physiological Review, 2004.

Accumulated amyloid-beta peptide and hyperphosphorylated tau protein: relationship and links in Alzheimer's disease. - J Alzheimers Dis., 2009.

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RESEARCH ARTICLE | ALZHEIMER'S DISEASE

β -amyloid redirects norepinephrine signaling to activate the pathogenic GSK3 β /tau cascade

 Fang Zhang^{1,*},  Mary Gannon^{1,*}, Yunjia Chen¹,  Shun Yan²,  Sixue Zhang³,  Wendy Feng¹,  Jiahui Tao¹,  Bingdong Sha¹, Zhenghui...

+ See all authors and affiliations

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Vol. 12, Issue 526, eaay6931
DOI: 10.1126/scitranslmed.aay6931

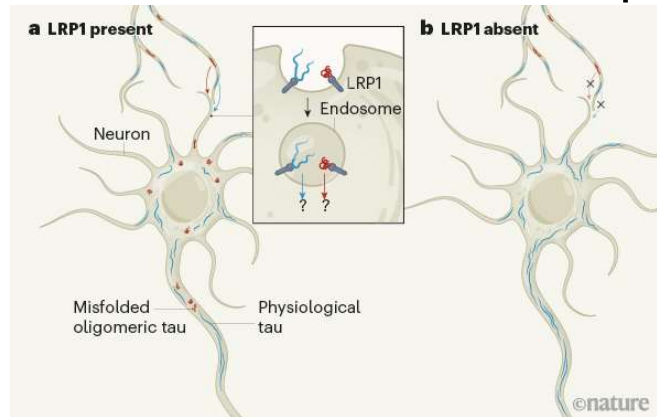
The Noradrenergic link

Norepinephrine (NE) and the noradrenergic system play a main role in cognition and noradrenergic changes have been documented in patients with Alzheimer's disease (AD). However, the role of NE in AD has not been completely elucidated. Here, Zhang *et al.* show that β -amyloid oligomers are allosteric ligands of the α 2A adrenergic receptor (α 2AAR) and modulate NE signaling, redirecting the pathway toward GSK3 β activation and subsequent tau hyperphosphorylation. Blocking α 2AAR reduced tau phosphorylation and ameliorated pathological and cognitive abnormalities in AD mouse models.

GSK3 β =Glycogen synthase kinase 3 beta, also implicated in bipolar disorder

<https://stm.sciencemag.org/content/12/526/eaay6931>

Misfolded forms of tau can spread



- 🔥 •Tau-associated forms of dementia (tauopathies) involve propagation of misfolded tau between neurons in a network
- LRP1** is a member of the cell surface LDL receptor family
- 🔥 •normal tau and misfolded tau oligomers are ligands for LRP1
- 🔥 •Binding leads to internalization of the tau
- 🔥 •Misfolded tau can act as a template to cause misfolding of normal tau

<https://www.nature.com/articles/s41586-020-2156-5>

<https://www.nature.com/articles/d41586-020-00874-z>

Biochemical changes in AD

Cholinergic hypothesis

- 🔥 loss of ACh-containing neurons and target neurons
- 🔥 – nucleus basalis of Maynert in particular
 - provides cholinergic neurons to cerebral cortex
- effects not restricted to cholinergic fibers
 - 5-HT, glutamate, neuropeptides.

Excitotoxic hypothesis

- 🔥 involves Ca^{2+} entry through NMDA glutamate receptors.

Vascular damage in AD

Chronic hypoperfusion hypothesis

- 🔥 Amyloid plaque formation in cerebral blood vessel endothelium
- 🔥 reduction in number of cerebral microvessels
- 🔥 significant aberrations in capillary walls
 - decreased capillary diameters and increased capillary densities
 - agrin deposition in capillaries suggestive of the thinning and fragmentation of the basal lamina
- atrophy of smooth muscle cells in cerebral vessels and attenuation of capillary endothelium resulting in the rupture of the vessel wall and intracerebral bleeding
- Resting CBF is reduced and the increase in CBF stimulated by neural activity is attenuated.

CBF = cerebral blood flow

CBF = cerebral blood flow

Vascular dementia

- 🔥 second most common form of dementia >65yo
- 🔥 symptoms similar to AD
 - result of interrupted blood flow to brain
 - post-stroke
 - multi-infarct (MID)
 - silent strokes: if stroke affects a small area, there may be no symptoms of a stroke
 - 🔥 – Over time, as more areas of the brain are damaged, the symptoms of MID begin to appear
 - larger strokes that have clear effects on strength, sensation, or other neurologic function also can lead to MID.

<http://www.nlm.nih.gov/medlineplus/ency/article/000746.htm>



Vascular dementia

- Patient may have past history of:
 - heart attacks
 - hypertension
 - high cholesterol
 - atherosclerosis
 - diabetes
 - other risk factors for heart disease are often present.

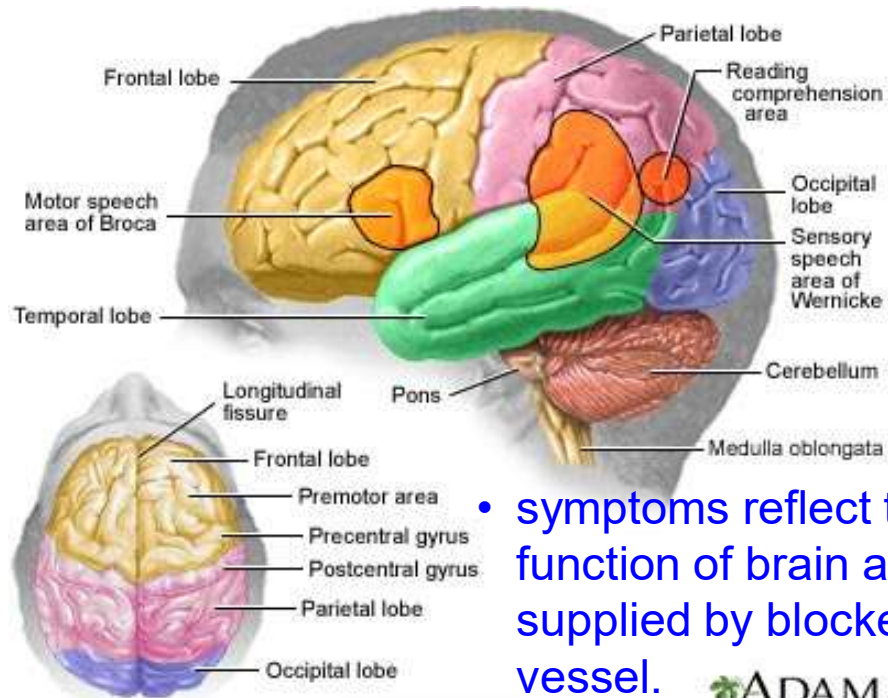
Vascular dementia is widely considered the second most common type of dementia. It develops when impaired blood flow to parts of the brain deprives cells of food and oxygen. The diagnosis may be clearest when symptoms appear soon after a single major stroke blocks a large blood vessel and disrupts the blood supply to a significant portion of the brain. This situation is sometimes called “post-stroke dementia.”

There is also a form in which a series of very small strokes, or infarcts, block small blood vessels. Individually, these strokes do not cause major symptoms, but over time their combined effect becomes noticeable. This type is referred to as vascular cognitive impairment (VCI) or multi-infarct dementia.

Symptoms of vascular dementia can vary, depending on the specific brain areas deprived of blood. Impairment may occur in “steps,” where there is a fairly sudden, noticeable change in function, rather than the slow, steady decline usually seen in Alzheimer’s disease.

The person may have a past history of heart attacks. High blood pressure, high cholesterol, hardening of the arteries, diabetes, or other risk factors for heart disease are often present.

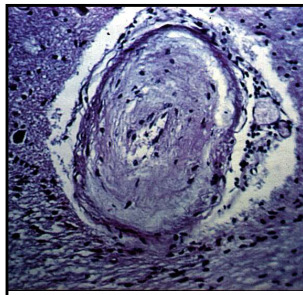
Vascular dementia



- symptoms reflect the function of brain area supplied by blocked vessel.

ADAM.

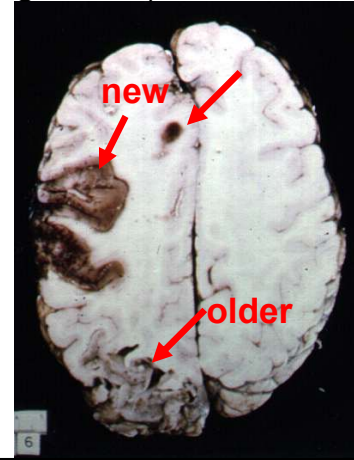
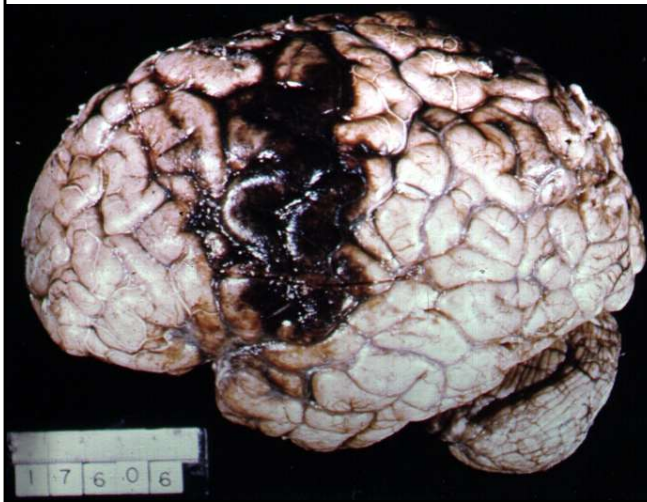
<http://www.clivir.com/lessons/show/vascular-dementia.html>



Arteriosclerosis of small blood vessel (small scale)

Vascular dementia (multi-infarct)

Hemorrhagic infarcts (large scale)



<http://www.urmc.rochester.edu/neuroslides/8-28.html>

Arteriosclerosis of small blood vessel: This is a trichrome stain of a small artery within the deep substance of the brain that shows marked thickening of its wall due to atherosclerosis. Arteriosclerotic degeneration with thrombosis involving small arteries in the central nervous system is the cause of lacunar infarcts, and may be associated with hypertension and to a lesser extent diabetes.

Hemorrhagic infarct, recent: The dark-colored convolutions ascending from the Sylvian fissure over the convexity of the hemisphere contain areas of hemorrhagic infarction. (Some of the vessels on the external surface are unusually prominent because of post-mortem injection of opaque material).

Hemorrhagic infarcts, recent, cut surface: This is a section cut horizontally which means that frontal lobe tissue lies at the upper part of the photograph and posterior parietal at the bottom. There are three areas of interest. The first is a recent (days old) hemorrhagic infarct in the mid portion of the lateral aspect of the left hemisphere where cortex and immediately adjacent subcortical white matter are discolored. The infarct does not involve the centrum semiovale. The second area is near the midline in frontal white matter where there is a small focus of recent hemorrhagic infarction. The third area is at the bottom of the photograph in the left posterior parietal region, where there is also a hemorrhagic infarct. This infarct has a granular appearance that indicates that it is a few weeks old.

Traumatic Brain Injury

- 🔥 **FALLS.** Leading cause of traumatic brain injury for all ages, and >75 y.o. are most likely to suffer hospitalization and death related to traumatic brain injury
- 🔥 Repeated TBIs (mild or severe) linked to CTE
 - no evidence that a single mild traumatic brain injury (TBI) increases dementia risk
- 🔥
 - within hours of severe TBI, beta amyloid levels increased in brain
 - Dementia (if present) characterized by tau deposits
 - Beta amyloid deposits may also be present
- 🔥 Permanent damage can occur even if no loss of consciousness
 - Treatment strategies for dementia are the same whether or not patient has TBI history
- CTE=Chronic Traumatic Encephalopathy

<http://www.alz.org/dementia/traumatic-brain-injury-head-trauma-symptoms.asp>

Older adults with a history of moderate traumatic brain injury had a 2.3 times greater risk of developing Alzheimer's than seniors with no history of head injury, and those with a history of severe traumatic brain injury had a 4.5 times greater risk.

Emerging evidence suggests that repeated mild traumatic brain injuries, such as those that can occur in sports like American football, hockey and soccer, may be linked to a greater risk of a type of dementia called chronic traumatic encephalopathy. Research has shown that boxers have an increased risk of chronic traumatic encephalopathy.

Current research on how traumatic brain injury changes brain chemistry indicates a relationship between traumatic brain injury and [hallmark protein abnormalities](#) (beta-amyloid and tau) linked to Alzheimer's.

Some research suggests that traumatic brain injury may be more likely to cause dementia in individuals who have a variation of the [gene for apolipoprotein E \(APOE\)](#) called APOE-e4. More research is needed to understand the link between APOE-e4 and dementia risk in those who've had a brain injury.

🔥 FDA Approved drugs for AD

Treat symptoms

ACh-esterase inhibitors

Donepezil (Aricept®)
Galantamine (Razadyne®)
Benzgalantamine (Zunveyl®) approved 2024
Rivastigmine (Exelon®)

NMDA R. antagonist

Memantine (Namenda®)

Disease modifying

Biologics (target/remove amyloid)

Aducanumab (Aduhelm®) discontinued 2024
Lecanemab (Leqembi®) approved 2023
Donanemab (Kisunla®) approved 2024

More about
treatments at
end of lecture!

<https://www.alz.org/alzheimers-dementia/treatments/medications-for-memory>

According to Biogen, aducanumab was discontinued in November 2024 in order for the company "to reprioritize its resources in Alzheimer's disease." Aducanumab was not discontinued for reasons related to safety or efficacy.

Alzheimer's disease - drugs

Acetylcholinesterase inhibitors

- mild to moderate AD, severe AD
- 🔥 may delay or prevent symptoms from becoming worse for a limited time
- 🔥 may help control some behavioral symptoms
- 🔥 increases effect of remaining cholinergic afferents
- may reduce A-beta₁₋₄₂ by stimulating antibody production

Neurosci Lett. 2010 Dec 17;486(3):193-6. Epub 2010 Oct 1. *Cholinesterase inhibitor use is associated with increased plasma levels of anti-Abeta 1-42 antibodies in Alzheimer's disease patients.* Conti E, Galimberti G, Tremolizzo L, Masetto A, Cereda D, Zanchi C, Piazza F, Casati M, Isella V, Appollonio I, Ferrarese C.

Donepezil

- approved for symptomatic management
 - For mild to severe AD
- 🔥 does not alter long term prognosis
 - improves global function, cognition, behavior
 - delays time to institutionalization
- higher CNS affinity → fewer PNS side effects
- dose 1/day, 100% oral bioavailability
- 🔥 metabolism by cytochrome P450

Donepezil
Aricept® | Aricept® ODTPrint

Aricept Tablet (Tab 5 mg)

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m

Classification:

Neurological Agents

Alzheimer's Agents

Cholinesterase inhibitors

Description: Donepezil was the first piperidine-type reversible cholinesterase (ChE) inhibitor to receive approval by the FDA. Donepezil is approved for the symptomatic management of mild to severe Alzheimer's disease. Prior to approval of donepezil, tacrine, an acridine-type cholinesterase (ChE) inhibitor, was the only agent approved for this disease, however, the acridine-type ChEs are associated with hepatotoxicity. Donepezil has not been associated with hepatotoxicity.^[24743] Donepezil has limited peripheral adverse reactions because of its greater affinity for CNS cholinesterase than tacrine. The long elimination half-life of donepezil allows for once daily dosing versus tacrine which is typically dosed four times daily. Clinical efficacy of donepezil in Alzheimer's disease is similar to other available ChE inhibitors (e.g., rivastigmine and tacrine) based on the Clinical Global Impression of Change (CGIC) scale and the Alzheimer's Disease Assessment Scale-Cognitive Subscale score (ADAS-cog). Donepezil does not alter the long-term prognosis of Alzheimer's disease, but delays the time until institutionalization of the patient, which may be cost-effective. In addition, one large multicenter study showed that the use of donepezil improved global function, cognition and behavior in those patients with moderate to severely advanced Alzheimer's disease (i.e., sMMSE score of 5–17) who did not yet require total nursing care.^[26898] In addition to head-to-head studies against other ChE-inhibitors, phase III trials of donepezil for Lewy Body dementia (LBD) and mild cognitive impairment (MCI) are in progress. Other clinical trials are ongoing to establish the safety, efficacy, and tolerability of donepezil in vascular dementia. Efficacy in severe Alzheimer's was established in a double-blind, controlled clinical trial involving 248 subjects.^[29640] Improvement in cognitive function compared to placebo was measured by the Severe Impairment Battery (SIB) instrument and overall function was measured by a caregiver assessment using the Modified Alzheimer's Disease Cooperative Study Activities of Daily Living Inventory for Severe Alzheimer's Disease (ADCS-ADL-severe). In a separate study, donepezil 10 mg/day was significantly better than placebo in treating severe Alzheimer's as assessed by the SIB and Clinician's Interview Based Impression of Change (CIBIC) plus.^[29640] In the same study, the 5 mg/day dose was found statistically superior to placebo on the SIB but not the CIBIC plus. Donepezil was approved by the FDA for mild to moderate Alzheimer's disease in November 1996. In October 2004, a rapid disintegration tablet (Aricept ODT) and an oral solution were approved by the FDA. In October 2006, the drug received approval for severe Alzheimer's disease dementia.

Mechanism of Action: Patients with Alzheimer's disease show behavioral consequences (e.g., decline in memory and learning) that are partially related to cholinergic deficits. CNS structural defects noted on biopsy or postmortem exam include cholinergic lesions in the nuclei projecting from the forebrain nucleus up to the cerebral cortex and the hippocampus, which is the specific region

involved with the function of memory. The cholinergic system is known to be important in attentional processing and as a modulator of excitatory amino acid (EAA) neurotransmission. Although there is presently no 'cure' for Alzheimer's disease, therapy with cholinesterase inhibitors is designed to offset the loss of presynaptic cholinergic function and slow the decline of memory and the ability to perform functions of daily living. This mechanism requires that intact cholinergic neurons be present. As Alzheimer's disease progresses, fewer intact cholinergic neurons remain, and cholinesterase inhibitors become less effective. There is considerable evidence indicating that, as in Alzheimer's disease, the central cholinergic system is also impaired in vascular dementia (VaD) and in patients with Alzheimer's disease with cerebrovascular disease ('mixed' dementia), as well as other conditions.

Donepezil selectively inhibits acetylcholinesterase, the enzyme responsible for the destruction of acetylcholine, and improves the availability of acetylcholine. Donepezil binds to AChE via hydrogen bonding and is easily hydrolyzed by body water, thus the duration of enzyme inhibition at the receptor level is very short, and referred to as 'reversible'. However, donepezil's long half-life provides a long duration of drug availability for binding at the receptor sites. Donepezil has much greater affinity for acetylcholinesterase (AChE) in the CNS than for butylcholinesterase (BChE) in the periphery, unlike the organophosphates, acridines, carbamates, physostigmine, and the quaternary ammonium anti-ChEs (amibenonium, neostigmine, pyridostigmine) which have similar affinity for both enzymes. There is no evidence to suggest that the underlying disease process of dementia is affected by administration of donepezil.

Pharmacokinetics: Donepezil is administered orally. After multiple dosing, donepezil accumulates in plasma by 4—7 fold and steady state is reached within 15 days. Donepezil is approximately 96% bound to human plasma proteins, primarily to albumin (about 75%) and alpha1-acid glycoprotein (about 21%) over the concentration range of 2—1000 ng/ml.

Donepezil is metabolized to four major metabolites, two of which are known to be active, and several minor metabolites. Metabolism occurs via hepatic cytochrome P450 isoenzymes 2D6 and 3A4 and by glucuronidation. Approximately 57% and 15% of an administered dose is excreted in the urine and feces, respectively, as metabolites. About 17% of a donepezil dose is recovered in urine as unchanged drug. Elimination half-life is about 70 hours.

•Route-Specific Pharmacokinetics

Oral Route

Following oral administration, donepezil is well absorbed with a relative bioavailability of 100%. The rate and extent of absorption of the tablet are not affected by administration with food. The effects of food on the absorption of the orally disintegrating tablet (ODT) have not been formally studied, but are expected to be minimal. Therefore, donepezil may be taken without regard to meals. With the exception of the ODT formulation, donepezil tablets should be swallowed whole. It has been noted by the manufacturer that crushing or chewing the 23 mg tablet may increase its rate of absorption. The kinetics of donepezil are linear over a range of 1 mg to 10 mg given once daily. Peak plasma concentrations of the 10 mg and 23 mg tablet are achieved in about 3 hours and 8 hours, respectively. Peak plasma concentrations following administration of the 23 mg tablet are almost 2-fold higher than those occurring with the 10 mg tablet. Steady state is achieved about 15 days after routine administration. [\[29640\]](#)

•Special Populations

Hepatic Impairment

In patients with stable alcoholic cirrhosis, donepezil clearance is reduced by roughly 20%.

Renal Impairment

The influence of renal impairment ($\text{CrCl} < 22 \text{ ml/min}$) on donepezil elimination has only been assessed in a limited number of patients; pharmacokinetics of donepezil do not appear to be appreciably altered in moderate or severe renal impairment.

Elderly

Population pharmacokinetic analysis suggests that the clearance of donepezil decreases with increasing age. Compared to 65-year old subjects, 90-year old subjects had a 17% decrease in clearance and 40-year old subjects had a 33% increase in clearance. The clinical significance of this finding is unknown. [\[29640\]](#)

Gender Differences

Donepezil pharmacokinetics do not appear to be influenced by gender.

Ethnic Differences

Donepezil pharmacokinetics do not appear to be influenced by race.

Other

Body Weight

Over a range of body weights from 50 kg to 110 kg, the clearance of donepezil increased from 7.77 L/hr to 14.04 L/hr. The value for 70 kg individuals was 10 L/hr.

References

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[Revised 9/2/2010 7:00:00 PM]

Galantamine

- natural alkaloid derived from daffodil
- 🔥 approved for mild to moderate AD
 - viable 1st line treatment
- 🔥 may also be beneficial in treating symptoms of vascular dementia and Dementia with Lewy bodies
- multiple daily dosages and a slow dosage titration to limit GI side effects
- partial metabolism via CYP450, renal excretion

Benzgalantamine

- Prodrug of galantamine

Galantamine
Razadyne™ | Razadyne™ER | Reminyl™ ER | Reminyl®Print

Galantamine Extended-Release Capsule (Capsule, Extended Release 16 mg)

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Classification:

Neurological Agents

Alzheimer's Agents

Cholinesterase inhibitors

Description: Galantamine, also known as galanthamine, is a cholinesterase (ChE) inhibitor. The drug is a natural alkaloid originating from the bulbs of the common snowdrop flower, also known as the daffodil or *Galanthus nivalis*. Galantamine is FDA approved for the treatment of mild to moderate Alzheimer's disease. Overall, galantamine appears to be a viable first-line treatment option for Alzheimer's disease; the drug improves cognitive functioning as assessed by the Alzheimer's Disease Assessment Scale-Cognitive subscale (ADAS-cog/11).^[26345] Similar to other ChE-inhibitors, the drug does not alter the progressive degenerative process of the disease. Limited data suggest that galantamine may also be beneficial in treating symptoms of vascular dementia and Dementia with Lewy bodies (DLB).^[41639] ^[41646] Galantamine has not been shown to be effective in mild cognitive impairment (MCI), a precursor of Alzheimer's disease; no slowing of progression to dementia was noted in data from two large (n = 2000), multicenter, international, 2-year long, placebo-controlled clinical trials. In addition, mortality data from these MCI trials trended toward a higher relative mortality rate in patients treated with galantamine compared to placebo; additional analysis is needed to determine whether galantamine truly had an adverse effect on mortality.^[31236] Galantamine has been used in humans for over 30 years as an anesthetic agent in anesthesia in countries outside the US. Also outside of the US, where the drug is known as Nivalin, galantamine has been used off-label for other conditions including glaucoma, muscular dystrophy, myasthenia gravis, and neuritis. Compared to donepezil and similar to rivastigmine, galantamine requires multiple daily dosages and a slow dosage titration to limit GI side effects. Hepatotoxicity has not been noted with galantamine as it has with tacrine. Galantamine tablets were approved by the FDA in February 2001 under the name Reminyl; an oral solution was approved in June 2001. The Reminyl brand name was changed to Razadyne in the U.S. to help avoid dispensing errors due to brand name confusion with other marketed drugs. Extended-release capsules (Razadyne ER) for once daily dosing were FDA-approved in December 2004.

Mechanism of Action: Patients with Alzheimer's disease show behavioral consequences (e.g., decline in memory and learning) that are partially related to cholinergic deficits. CNS structural defects noted on biopsy or postmortem exam include cholinergic lesions in the nuclei projecting from the forebrain nucleus up to the cerebral cortex and the hippocampus, which is the specific region involved with the function of memory. The cholinergic system is known to be important in attentional processing and as a modulator of excitatory amino acid (EAA) neurotransmission. Although there is presently no 'cure' for Alzheimer's disease, therapy with cholinesterase inhibitors is designed to offset the loss of presynaptic cholinergic function and slow the decline of memory and the ability to perform functions of daily living. This mechanism requires that intact cholinergic neurons be present. As Alzheimer's

disease progresses, fewer intact cholinergic neurons remain, and cholinesterase inhibitors become less effective. There is considerable evidence indicating that, as in Alzheimer's disease, the central cholinergic system is also impaired in vascular dementia (VaD) and in patients with Alzheimer's disease with cerebrovascular disease ('mixed' dementia), as well as other conditions.

Galantamine is a tertiary alkaloid, and similar to tacrine, donepezil, and physostigmine, the drug is a reversible inhibitor of acetylcholinesterase (AChE). A single 10 mg oral dose resulted in an average maximum 39% inhibition of AChE at 60 minutes in erythrocytes from healthy volunteers. Since galantamine is tertiary in structure, it penetrates the CNS and can inhibit brain AChE. Galantamine also enhances the effect of acetylcholine on nicotinic receptors, likely through binding to an allosteric site of the receptor. The dual action is a point of differentiation from the other cholinesterase inhibitors. However, human data providing clinically significant evidence of the drug's actions on nicotinic receptors is not yet available. There is no evidence to suggest that the underlying disease process of dementia is affected by administration of galantamine.

Pharmacokinetics: Galantamine is administered orally. The plasma protein binding of galantamine is 18% at therapeutically relevant concentrations. In whole blood, galantamine is mainly distributed to blood cells (roughly 53%).

Galantamine does not appear to be a significant inhibitor of the hepatic CYP450 microsomal enzymes; however, it is partially metabolized via this system. Multiple metabolic pathways and renal excretion are involved in the elimination of galantamine so no single pathway appears predominant. Hepatic isoenzymes CYP2D6 and CYP3A4 are involved in the formation of the metabolites O-desmethylgalantamine and N-oxide-galantamine, respectively. O-demethylation, mediated by CYP2D6, is greater in extensive metabolizers of CYP2D6 than in poor metabolizers. Population pharmacokinetic analysis indicate that there is a 25% decrease in median clearance in poor metabolizers compared to extensive metabolizers. However, dosage adjustment is not necessary in patients identified as poor metabolizers as the dose of drug is individually titrated to tolerability. The elimination half-life averages 7 hours. Approximately 20% of a dose is excreted through the kidneys in 24 hours as unchanged galantamine. In studies of radiolabeled oral galantamine, unchanged galantamine and its glucuronide accounted for most of the plasma radioactivity. By 7 days, 93—99% of the total radioactivity was recovered, with about 95% in urine and about 5% in the feces.

•Route-Specific Pharmacokinetics

Oral Route

Galantamine is rapidly and completely absorbed following oral administration with time to peak concentration about 1 hour. Bioavailability of the immediate-release tablet is the same as an oral solution; bioavailability is roughly 90%. The rate of absorption is delayed by administration with food; however, the extent of absorption is not affected. Administration with food is recommended to limit drug intolerance.

•Special Populations

Hepatic Impairment

After a single 4 mg dose in eight subjects with moderate hepatic impairment (i.e., Child-Pugh score 7—9), the clearance of galantamine was decreased by about 25% compared to healthy volunteers. The effects on metabolism in patients with severe hepatic impairment (i.e., Child-Pugh score 10—15) are unknown. [\[32342\]](#)

Renal Impairment

Following a single 8 mg dose of galantamine, AUC increased by 37% and 67% in moderate and severely renally-impaired patients, respectively, compared to normal adults. The drug should be avoided in those with a creatinine clearance < 9 ml/min.

Other

Alzheimer's Disease

Data from clinical trials in patients with Alzheimer's disease indicate that galantamine serum concentrations are 30—40% higher than in healthy young adults.

References

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[Revised 9/13/2010 5:34:00 PM]

Rivastigmine

- approved for
 - mild to moderate AD (2000) ,
 - mild to moderate dementia in Parkinson's disease (2006)
- Clinical efficacy similar to donepezil
- 🔥 Not metab. by cyt.P450; low protein-binding
 - drug interactions less likely
- 🔥 metabolized by cholinesterase at CNS sites
- side effects mostly GI
 - available in a patch with fewer side effects, which is also approved for severe AD

Rivastigmine Exelon®Print

Exelon Solution (Sol 2 mg/ml)

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Classification:

Neurological Agents

Alzheimer's Agents

Cholinesterase inhibitors

Description: Rivastigmine is a CNS-selective carbamate-type cholinesterase (ChE) inhibitor used for the treatment of mild to moderate Alzheimer's disease. The drug is structurally distinct from tacrine and donepezil. Rivastigmine has been studied in > 3300 patients and is approved in over 40 countries worldwide. Clinical efficacy of rivastigmine is similar to donepezil and tacrine based on the Clinical Global Impression of Change (CGIC) scale and Alzheimer's Disease Assessment Scale-Cognitive Subscale score (ADAS-cog) at 6 months of treatment. Like other cholinergic agents, rivastigmine does not alter the long-term prognosis of Alzheimer's disease, but probably delays the time until institutionalization of the patient. Rivastigmine is not metabolized by hepatic cytochrome P450 oxidative isoenzymes and it exhibits low protein-binding. Thus, drug interactions with rivastigmine are less likely to occur than with other ChE inhibitors (i.e., donepezil and tacrine). However, like other ChE inhibitors, rivastigmine dosage requires careful titration to limit GI side effects. In the US, head-to-head studies against other ChE-inhibitors are planned; the manufacturer will also pursue phase III trials for Lewy Body dementia (LBD), vascular dementia [26844], and mild cognitive impairment (MCI). The FDA initially approved rivastigmine capsules and oral solution for mild to moderate Alzheimer's disease in April 2000. It was approved for mild to moderate dementia associated with Parkinson's disease in June 2006. Rivastigmine is the first approved agent in the US for dementia associated with Parkinson's disease; approval was based on the results from the EXPRESS study. [32042] Rivastigmine also has approval for this indication by the European Union (EU). Results from a large, randomized, comparison trial of rivastigmine once daily transdermal (20-cm² or 10-cm²) to twice daily oral (12 mg/day) capsules and placebo found that the patch was as effective as rivastigmine capsules for the treatment of Alzheimer's disease. [32788] Both formulations were superior to placebo. The side effect profile was similar between the 20-cm² patch and oral treatment. The 10-cm² patch was as effective as oral rivastigmine but was associated with significantly less side effects. In July 2007, the FDA approved rivastigmine transdermal system (Exelon® patch) for mild to moderate Alzheimer's disease or mild to moderate dementia associated with Parkinson's disease. In September 2007, the EU approved the Exelon® patch for mild to moderately severe Alzheimer's disease.

Mechanism of Action: Patients with Alzheimer's disease show behavioral consequences (e.g., decline in memory and learning) that are partially related to cholinergic deficits. CNS structural defects noted on biopsy or postmortem exam include cholinergic lesions in the nuclei projecting from the forebrain nucleus up to the cerebral cortex and the hippocampus, which is the specific region involved with the function of memory. The cholinergic system is known to be important in attentional processing and as a

modulator of excitatory amino acid (EAA) neurotransmission. Although not a cure, therapy with cholinesterase inhibitors is designed to offset the loss of presynaptic cholinergic function and slow the decline of memory and the ability to perform functions of daily living. This mechanism requires that intact cholinergic neurons be present. As dementia progresses, fewer intact cholinergic neurons remain, and cholinesterase inhibitors become less effective. There is considerable evidence indicating that, as in Alzheimer's disease, the central cholinergic system is also impaired in vascular dementia (VaD) and in patients with Alzheimer's disease with cerebrovascular disease ('mixed' dementia), as well as other conditions.

Rivastigmine is a potent, selective inhibitor of brain acetylcholinesterase (AChE) and butylcholinesterase (BChE). In animal studies, rivastigmine produced a 10-fold greater inhibition of AChE in the hippocampus and cortex than its effects on BChE and AChE in the heart, skeletal muscle, and other peripheral tissues, which may explain its relatively low incidence of peripheral cholinergic side effects with appropriate dose titration. The selective effect in the cortex and hippocampus may be due to its preferential inhibition of the G1 form of the acetylcholinesterase enzyme, which is present in relatively higher concentrations in these brain areas. Unlike tacrine, donepezil, galantamine and physostigmine, which are reversible inhibitors of cholinesterase, and metrifonate, which is considered to be an irreversible inhibitor, rivastigmine is considered a pseudo-irreversible inhibitor of AChE. Rivastigmine binds to the esteratic site of the acetylcholinesterase enzyme but dissociates much more slowly than acetylcholine. This 'pseudo-irreversible' action explains why the cholinesterase inhibition of rivastigmine in the brain lasts much longer (average 10 hours) than the short plasma half-life of the drug would predict. There is no evidence to suggest that the underlying disease process is affected by administration of rivastigmine.

Pharmacokinetics:

Rivastigmine is administered orally or transdermally. Intersubject variability in rivastigmine exposure is lower for the transdermal system (43%–49%) than the oral formulation (73–103%). Rivastigmine is weakly bound to plasma proteins (approximately 40%). It readily crosses the blood brain barrier and is widely distributed. Approximately 50% of the drug load is released from the transdermal system over 24 hours.

Hepatic cytochrome P450 isoenzymes (CYP450) are minimally involved in rivastigmine metabolism. Rivastigmine is rapidly and extensively metabolized primarily at CNS receptor sites via cholinesterase, which mediates hydrolysis to the decarbamylated phenolic metabolite (i.e., ZNS 114–666). This metabolite is detected within 2 hours of rivastigmine administration. In vitro, the decarbamylated metabolite shows minimal inhibition of acetylcholinesterase (< 10%). The ZNN–666 metabolite is N-demethylated or sulfated in the liver, but is of no therapeutic consequence. Consistent with these observations is the finding that no drug interactions relating to CYP450 have been observed in humans. The plasma half-lives of rivastigmine and the ZNN–666 metabolite are roughly 1 hour and 2 hours, respectively; however, the cholinesterase inhibition in the CNS lasts much longer (average 10 hours) than the short plasma half-life would predict. This is due to the fact that when rivastigmine's phenolic ZNN–666 metabolite is formed, it leaves behind a carbamate moiety that stays attached to the AChE receptor for up to 10 hours, which prevents the hydrolysis of ACh.

Renal excretion of the ZNN–666 metabolite is the major route of elimination; unchanged rivastigmine is not found in the urine. Renal elimination is essentially complete (> 90%) within 24 hours. Less than 1% of the administered dose is excreted in the feces. Although patients with Alzheimer's disease demonstrate 30–50% higher plasma concentrations of rivastigmine and its decarbamylated metabolite than do healthy elderly patients, there is no evidence of drug accumulation, which is consistent with the short plasma half-life.

•Route-Specific Pharmacokinetics

Oral Route

Following oral administration, it is rapidly and completely absorbed and peak plasma concentrations are reached in approximately 1 hour. Absolute bioavailability after a 3-mg oral dose is 36%, indicating a significant first-pass effect. Oral administration with food delays absorption and lowers C_{max} by roughly 30%, but increases the AUC by approximately 30%. Thus, oral rivastigmine should be taken with food to enhance bioavailability and to increase tolerability of the medication.

Topical Route

Following transdermal application, absorption begins within 30 minutes to 1 hour and peak plasma concentrations are typically reached in 8 hours (range: 8–16 hours). With transdermal application, trough levels are about 60–80% of peak levels at steady state. Body weight affects rivastigmine exposure; steady state concentrations are approximately doubled in a patient weighing 35 kg compared to 65 kg. Following a transdermal dose of 9.5 mg/24 hours, drug exposure is similar to an oral dose of 6 mg twice daily. Approximately 50% of the drug load is released from the transdermal system over 24 hours.

•Special Populations

Hepatic Impairment

In patients with mild to moderate hepatic impairment, the rivastigmine AUC was roughly 2.3-fold higher than in healthy subjects, but the changes may be attributable to changes in plasma protein levels, fluid balance or other non-metabolic conditions that occur in these subjects. No dosage adjustments are specifically recommended. Rivastigmine has not been studied in patients with severe hepatic impairment.

Renal Impairment

The metabolism and excretion of rivastigmine may be altered to some degree by the presence of renal impairment. In patients with moderate to severe renal impairment, no differences were found in the elimination half-life of rivastigmine; the metabolite's half-life was slightly prolonged. The AUC of rivastigmine was 1.4-fold higher and the metabolite's AUC was 0.8-fold lower than that of healthy volunteers. Based on these parameters, no specific dosage adjustments are recommended.

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[Revised 10/4/2010 8:37:00 PM]

Drug Information Provided by
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Memantine



NMDA R antagonist (use-dependent non-competetive)

- low affinity, prevents excitotoxicity, LTP o.k. (learning and memory)
- partial affinity for nACh R



approved for moderate-severe AD (2003)



Reduction of care dependence



Neuroprotective – slows disease progression

- Very good tolerability (side effect frequency similar to placebo)
- Not metabolized by cytochrome P450 (Few drug interactions)
- partial hepatic metabolism, active renal secretion
 - elim. $\frac{1}{2}$ life 60-80 hours, clearance decreased in alkaline urine, increased in acid



Extended-release formulation approved (2010)

Memantine
Namenda XR | Namenda®Print

Namenda Tablet Titration Pak (Tablet 5;10 mg;mg)
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Classification:

Neurological Agents
 Alzheimer's Agents

Description: Memantine is an oral, non-competitive antagonist at N-methyl-D-aspartate (NMDA) receptors. Memantine is indicated for the treatment of moderate-to-severe Alzheimer's Disease (AD) and is being studied for treatment of mild-to-moderate vascular dementia. Pivotal placebo-controlled trials have been completed for the AD indication.^[27648] Memantine is the first therapy approved for the more advanced stages of AD when difficulty with daily activities, behavioral problems and caregiver burden become significant. Memantine may have neuroprotective effects due to a novel mechanism of action distinct from the currently available cholinergic therapies for AD. Memantine has been studied (n=404) in combination with donepezil for treatment of moderate to severe Alzheimer's disease and initial results have been positive for measures of cognition, activities of daily living, global outcome and behavior vs. memantine alone.^[27813] Memantine has also been investigated for dementia syndromes due to AIDS, cerebral ischemia and neuropathic pain associated with diabetes. Controlled phase III trial data have not been positive for the neuropathic pain use. Additionally, memantine has not been shown to be effective in the treatment of chronic phantom limb pain (PLP). In chronic PLP studies, oral memantine doses of 30 mg/day were associated with a higher incidence of serious adverse events versus placebo; the authors conclude that dosages > 30 mg/day may not be tolerable.^[27649] A small number of patients with acquired pendular nystagmus have responded to treatment with memantine, although further study of this potential treatment option is needed.^[32638] Memantine has been marketed in Germany since the 1980's for the management of dementia; the drug has been available for treatment of moderate to severe AD in the European Union since June 2002. The FDA approved memantine for the treatment of moderate to severe AD in October 2003. In November 2004 the FDA accepted the NDA filing of Forest Laboratories for memantine treatment of mild Alzheimer's disease (AD); FDA action is pending. In April 2005 an oral solution of memantine (2 mg/ml) was approved for use by the FDA. In June 2010, an extended-release capsule (Namenda XR) was approved by the FDA for the treatment of moderate to severe Alzheimer's disease.

Mechanism of Action: Glutamate activation at the NMDA receptor is needed for learning and memory processes in the brain. There is evidence that the chronic excitatory activity of the neurotransmitter L-glutamate may play a role in the pathogenesis of Alzheimer's disease and other neurological disorders. Glutamate, the dominant excitatory amino acid in the brain, stimulates NMDA receptors, which in turn leads to increased intraneuronal concentrations of calcium (Ca²⁺). Excessive glutamate (and therefore Ca²⁺) in the CNS can lead to neuronal damage. For example, persistently elevated synaptic glutamate levels during hypoxia and ischemia can lead to cell death due to excess Ca²⁺ influx via NMDA receptor channels. Under normal resting conditions, NMDA receptor channels do not allow chronic Ca²⁺ influx because they are blocked by magnesium (Mg²⁺).

Depolarization of the membrane allows the Ca²⁺ to bypass the Mg²⁺ blockade, and influx through the NMDA receptor channels.

Memantine is an antagonist at (N-methyl-D-aspartate) NMDA receptors; memantine has a low to moderate affinity for the NMDA receptor. Blockade of NMDA receptors by memantine slows the intracellular calcium accumulation and helps to prevent further nerve damage. A low affinity antagonist to NMDA-type receptors, such as memantine, may prevent excitatory amino acid neurotoxicity without interfering with the physiological actions of glutamate required for memory and learning. High affinity NMDA antagonists, such as ketamine and dextromethorphan, are associated with excessive psychotomimetic effects at dosages needed for dementia treatment.[\[27650\]](#)

Memantine does not affect the release of dopamine or serotonin, nor does it alter monoamine oxidase (MAO-A or B) or adenylate cyclase activity. *In-vitro* studies demonstrate that memantine lacks affinity for most serotonin receptor subtypes (except 5HT₃), muscarinic acetylcholine, α and β adrenergic, dopaminergic, histaminic, and glycine receptors. Memantine appears to have antagonist activity at the 5HT₃ receptor, with similar potency to that of the NMDA receptor.[\[31418\]](#) Memantine also has partial affinity for nicotinic acetylcholine receptors, with one-sixth to one-tenth the potency (relative to the NMDA receptor).[\[31418\]](#)

Pharmacokinetics: Memantine is administered orally as an immediate-release tablet, extended-release capsule, and oral solution. Protein binding, roughly 45%, is not a clinical concern. Memantine rapidly crosses the blood brain barrier and can be detected in the CSF within 30 minutes of an IV infusion. The volume of distribution is 9 to 11 L/kg suggesting extensive distribution into tissues. Steady-state therapeutic plasma concentrations (0.37—0.5 microM) are similar between patients with dementia and healthy controls.

Memantine undergoes partial hepatic metabolism. *In-vitro*, memantine shows minimal inhibition of the cytochrome P450 (CYP) enzyme system and minimal substrate interactions. Renal clearance is 182 ml/min and substantially exceeds the glomerular filtration rate, indicating active renal secretion. Increases in urinary pH may decrease the elimination of memantine, resulting in drug accumulation and potential toxicity. The clearance of memantine is reduced by about 80% under alkaline urine conditions at pH 8 (see Drug Interactions). Alternatively, acidification of the urine may increase the elimination of memantine. Memantine has an elimination half-life of 60—80 hours. About 48% of the dose is excreted in urine as unchanged drug or as the sum of parent drug and the N-glucuronide conjugate (74%).

•Route-Specific Pharmacokinetics

Oral Route

Memantine is rapidly and completely absorbed with 100% bioavailability. Food does not alter the extent of absorption of the immediate-release (IR) or extended-release (ER) formulation; therefore, the drug may be taken with or without food. There is no difference in the systemic exposure of extended-release memantine between swallowing the capsule intact or sprinkling the contents on applesauce. However, other methods of administration such as cutting, dividing, or chewing the capsules should be avoided.[\[41140\]](#) The pharmacokinetics of immediate-release memantine are linear in the range of 5—40 mg single oral doses. Peak concentrations of the IR and ER formulations occur in 3—7 hours and 9—12 hours, respectively. In a study comparing 28 mg/day of ER memantine to 10 mg twice daily of IR memantine, the extended-release memantine had a C_{max} and AUC that were 48% and 33% higher, respectively.[\[41140\]](#) The serum concentration of memantine is roughly double that found in the cerebrospinal fluid (CSF).

Intravenous Route

Memantine can be detected in the CSF within 30 minutes of an IV infusion.

•Special Populations

Hepatic Impairment

In a small pharmacokinetic study (n=16) evaluating single doses of memantine 20 mg in healthy patients compared to those with moderate hepatic impairment (Child-Pugh Class B, score 7-9), the elimination half-life increased by about 16% in the hepatic impairment group compared to the healthy subjects. No change in C_{max} or AUC was apparent between the two groups.[\[41140\]](#) According to the manufacturer, no dosage adjustments of memantine are needed in patients with mild to moderate hepatic impairment; however caution is advised when using the drug in those with severe hepatic impairment.

Renal Impairment

In a pharmacokinetic study evaluating patients with renal impairment, the mean AUC increased by 4%, 60%, and 115% in subjects with mild, moderate, and severe renal impairment, respectively, compared to healthy subjects. The terminal elimination half-life increased by 18%, 41%, and 95% in subjects with mild, moderate, and severe renal impairment, respectively. According to the manufacturer, no dosage adjustment is recommended for patients with mild and moderate renal impairment; however, the dosage should be reduced in those with severe renal impairment (see Dosage). In the elderly, renal impairment results in decreased clearance of memantine and increased AUC, corresponding with decreasing values of creatinine clearance (CrCl).

Elderly

Memantine pharmacokinetics do not appear to be affected by the age of the patient. In the elderly, renal impairment results in decreased clearance of memantine and increased AUC, corresponding with decreasing values of creatinine clearance (CrCl).

Gender Differences

Following multiple doses of memantine 20 mg/day, females had approximately a 45% higher exposure than males, but no differences were noted after adjustments for body weight.[\[41140\]](#)

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Drug Updates Story

Extended-release formulation of dementia drug Namenda approved

June 23, 2010

ST LOUIS (MD Consult) - On June 21, 2010, Forest Laboratories announced that the US Food and Drug Administration (FDA) has approved Namenda XR (memantine) for the treatment of moderate to severe dementia of the **Alzheimer's** type. *Namenda XR* is a 28-mg once-daily extended-release formulation of Namenda. *Namenda*, approved by the FDA in 2003, is an immediate-release formulation of memantine that is administered twice daily.

Namenda XR has a mechanism of action that focuses on the glutamate pathway, a target for the treatment of **Alzheimer's** disease.

The safety and efficacy of Namenda XR were established in a randomized, double-blind, placebo-controlled trial of 677 outpatients already receiving a cholinesterase inhibitor. Study results indicate that patients treated with Namenda XR experienced statistically significant benefits in cognition and clinical global status compared with patients given placebo.

In clinical trials, the most common adverse events occurring in at least 5% of patients treated with Namenda XR and at a greater frequency than patients treated with placebo were headache (6% vs 5%), diarrhea (5% vs 4%), and dizziness (5% vs 1%).

Namenda XR should be used with caution under conditions that raise urinary pH. Alkaline urine conditions may decrease the urinary elimination of memantine, resulting in increased plasma levels.

Namenda XR should be administered with caution to patients with severe hepatic impairment. A dosage reduction is recommended in patients with severe renal impairment.

Biologics

- **Lecanemab (Leqembi®)** approved 2023
 - Requires IV infusion q 2 weeks
- **Donanemab-azbt (Kisunla®)** approved 2024
 - Requires IV infusion q 4 weeks



Recombinant human mAb



Approved for mild cognitive impairment (MCI) or mild dementia if there is confirmation of elevated beta-amyloid in the brain



Binds to aggregated forms of Abeta

- Soluble oligomers and insoluble fibrils

- Does not bind Abeta monomers

- Side effects



ARIA “amyloid-related imaging abnormalities”

- brain edema, hemorrhages, hemosiderosis (iron deposition in neuronal tissue)
 - Headache, nausea, vision changes, falls

Lecanemab Slowed rate of cognitive decline by 27%

At-a-glance treatment chart

Generic	Brand	Approved For	Side Effects
donepezil	Aricept	All stages	Nausea, vomiting, loss of appetite and increased frequency of bowel movements.
galantamine	Razadyne	Mild to moderate	Nausea, vomiting, loss of appetite and increased frequency of bowel movements.
memantine	Namenda	Moderate to severe	Headache, constipation, confusion and dizziness.
rivastigmine	Exelon	Mild to moderate Patch also approved for severe	Nausea, vomiting, loss of appetite and increased frequency of bowel movements.
memantine + donepezil	Namzaric	Moderate to severe	Nausea, vomiting, loss of appetite, increased frequency of bowel movements, headache, constipation, confusion and dizziness.

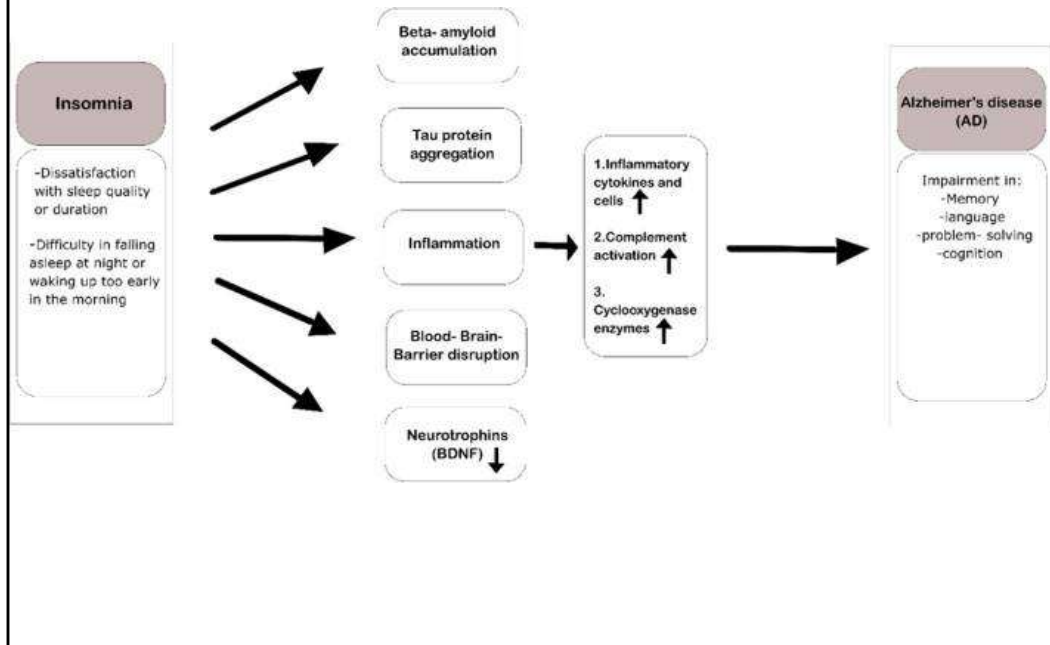
Other Treatment Trials

(not FDA approved)

- Abeta42 vaccination
 - Worked in mice
 - led to life-threatening complications, including meningoencephalitis, in a minority of human subjects
- Beta-, Gamma-secretase inhibitors
 - Controlled trials failed, possibly due to advanced stage of disease in subjects
- nSAIDs and Statins
 - may have a protective effect on dementia if used prior to the onset of disease but do not influence clinically symptomatic AD.
- Minocycline
 - Has multiple anti-inflammatory effects, reduces Abeta deposition
 - Showed no benefit in trials

Preventive Measures

🔥 Sleep is good.



<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7542374/>

The effect of insomnia on development of Alzheimer's disease

Preventive Measures

Good Diet is good.

- Accumulating evidence from numerous trials of defined diets:
 - Mediterranean
 - associated with better cognitive scores in 9 of 12 cross-sectional studies, 17 of 25 longitudinal studies, and 1 of 3 trials.
 - lower risk of AD in 1 case-control study and 6 of 8 longitudinal studies.
 - Dietary Approaches to Stop Hypertension (DASH)
 - associated with better cognitive function in 1 cross-sectional study, 2 of 5 longitudinal studies, and 1 trial.
 - Mediterranean-DASH Intervention for Neurodegenerative Delay (MIND)
 - Strongly associated with better cognitive scores in 1 cross-sectional study and 2 of 3 longitudinal studies.
 - lower AD risk in 1 longitudinal study.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6855954/>

The Mediterranean, Dietary Approaches to Stop Hypertension (DASH), and Mediterranean-DASH Intervention for Neurodegenerative Delay (MIND) Diets Are Associated with Less Cognitive Decline and a Lower Risk of Alzheimer's Disease—A Review

Van den Brink et al.

[Adv Nutr.](#) 2019 Nov; 10(6): 1040–1065.

Published online 2019 Jun 18. doi: [10.1093/advances/nmz054](https://doi.org/10.1093/advances/nmz054)

Preventive Measures

 Exercise is good.

- Increases plasma cathepsin B
- Increases plasma lactate and ketone bodies
- Increases BDNF
 - Increased adult hippocampal neurogenesis
 - Increased synaptic plasticity
 - Increased cognitive function
- Decreases neuroinflammation
- Reduces Amyloid beta load

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7542374/>

The effect of insomnia on development of Alzheimer's disease



Treating Behavioral Changes in AD

- In **early** stages, behavior and personality changes:
 - Irritability
 - Anxiety
 - Depression
- In **later** stages, other symptoms may include:
 - Anger
 - Agitation
 - Aggression
 - General emotional distress
 - Physical or verbal outbursts
 - Restlessness, pacing, shredding paper or tissues
 - Hallucinations
 - Delusions
 - Sleep disturbances
 - “Sundowning”.

Factors that may contribute to trouble sleeping and sundowning

- Mental and physical exhaustion from a full day of activities.
- Navigating a new or confusing environment.
- A mixed-up "internal body clock." The person living with Alzheimer's may feel tired during the day and awake at night.
- Low lighting can increase shadows, which may cause the person to become confused by what they see. They may experience hallucinations and become more agitated.
- Noticing stress or frustration in those around them may cause the person living with dementia to become stressed as well.
- Dreaming while sleeping can cause disorientation, including confusion about what's a dream and what's real.
- Less need for sleep, which is common among older adults.

Tips that may help manage sleep issues and sundowning

- Encourage the person living with dementia to get plenty of rest.
- Schedule activities such as doctor appointments, trips and bathing in the morning or early afternoon hours when the person living with dementia is more alert.
- Encourage a regular routine of waking up, eating meals and going to bed.
- When possible, spend time outside in the sunlight during the day.

- Make notes about what happens before sundowning events and try to identify triggers.
- Reduce stimulation during the evening hours. For example, avoid watching TV, doing chores or listening to loud music. These distractions may add to the person's confusion.
- Offer a larger meal at lunch and keep the evening meal lighter.
- Keep the home well lit in the evening to help reduce the person's confusion.
- Try to identify activities that are soothing to the person, such as listening to calming music, looking at photographs or watching a favorite movie.
- Take a walk with the person to help reduce their restlessness.
- Talk to the person's doctor about the best times of day for taking medication.
- Try to limit daytime naps if the person has trouble sleeping at night.
- Reduce or avoid alcohol, caffeine and nicotine, which can all affect the ability to sleep.

Treating Behavioral Changes in AD

Try non-pharmacologic approaches first:

- 🔥 Avoid triggering situations
 - Changes to familiar environment
 - Misperceived threats
- Advise caregivers in coping strategies:
 - 🔥 – Monitor personal comfort – pt may not be able to communicate that something is wrong
 - 🔥 – Avoid confrontation about mistaken facts
 - 🔥 – Redirect pt's attention
 - Create calm environment
 - 🔥 – Allow rest between stimulating events

Treating Behavioral Changes in AD



Brexpiprazole: approved for agitation in AD dementia (2023)

Suvorexant (DORA): approved for insomnia in AD (2020)

Off label drug usage:



Antidepressants for low mood and irritability:

- citalopram
- fluoxetine
- paroxetine
- sertraline
- trazodone



Anxiolytics for anxiety, restlessness, verbally disruptive behavior and resistance:

- lorazepam
- oxazepam



Antipsychotic medications for hallucinations, delusions, aggression, agitation, hostility and uncooperativeness:

- aripiprazole
- clozapine
- haloperidol
- olanzapine
- quetiapine
- risperidone
- ziprasidone

Non-cognitive symptoms (behavioral and psychological symptoms)

•Alzheimer's affects more than just memory and thinking. A person's quality of life may be impacted by a variety of behavioral and psychological symptoms that accompany dementia, such as [sleep disturbances](#), [agitation](#), [hallucinations and delusions](#). Some medications focus on treating these non-cognitive symptoms for a time, though it is important to try [non-drug strategies to manage behaviors](#) before adding medications.

The FDA has approved one drug to address symptoms of insomnia that has been tested in people living with dementia and one that treats agitation.

Orexin receptor antagonist

Prescribed to treat insomnia, this drug inhibits the activity of orexin, a type of neurotransmitter involved in the sleep-wake cycle:

Suvorexant (Belsomra®): approved for treatment of insomnia and has been shown in clinical trials to be effective for people living with mild to moderate Alzheimer's disease. Possible side effects include, but are not limited to: risk of impaired alertness and motor coordination (including impaired driving), worsening of depression or suicidal thinking, complex sleep behaviors (such as sleep-walking and sleep-driving), sleep paralysis and compromised respiratory function.

•**Atypical antipsychotics**

Atypical antipsychotics are a group of antipsychotic drugs that target the serotonin and dopamine chemical pathways in the brain. These drugs are largely used to treat schizophrenia and bipolar disorder and as add-on therapies for major depressive disorder. The FDA requires that all atypical antipsychotics carry a safety warning that the medication has been associated with an increased risk of death in older patients with dementia-related psychosis.

Many atypical antipsychotic medications are used "off-label" to treat dementia-related behaviors, and there is currently only one FDA-approved atypical antipsychotic to treat agitation associated with dementia due to Alzheimer's. It is important to try non-drug strategies to manage non-cognitive symptoms — like agitation — before adding medications.

Brexpiprazole (Rexulti®): approved for the treatment of agitation associated with dementia due to Alzheimer's disease. Possible side effects include, but are not limited to: weight gain, sleepiness, dizziness, common cold symptoms, and restlessness or feeling like you need to move. Warning for serious side effects: increased risk of death in older adults with dementia-related psychosis. Rexulti is not approved for the treatment of people with dementia-related psychosis without agitation that may happen with dementia due to Alzheimer's disease.

Treating Behavioral Changes in AD

- 🔥 Use drugs to target specific behavioral symptoms
 - Start with low dose of single drug, monitor side effects closely
- 🔥 **Caution (Black Box Warning):**
 - 🔥 Antipsychotics are associated with an **increased risk of stroke and death** in older adults with dementia.
 - 🔥 Not approved to treat dementia related psychosis
 - Sometimes used for this anyway. Only use if:
 - Behavioral symptoms are due to mania or psychosis
 - The symptoms present a danger to the person or others
 - The person is experiencing inconsolable or persistent distress, a significant decline in function or substantial difficulty receiving needed care.

See warning on FDA website:

<http://www.fda.gov/Drugs/DrugSafety/PostmarketDrugSafetyInformationforPatientsandProviders/ucm124830.htm>

Treating Sleep Disturbance in AD

- 🔥 Dual Orexin R antagonist (DORA) (suvorexant)
- Tricyclic antidepressants (nortriptyline, trazodone)
- Benzodiazepines (lorazepam, oxazepam)
- Sedative-hypnotics (zolpidem, zaleplon)
- Antipsychotics (risperidone, olanzapine, haloperidol)

- **Caution:**

- 🔥 – Sleep inducing meds are risky in patients with cognitive impairment
 - Increased falls and fractures
 - Confusion
 - Decline in self-care

Resources



<https://www.Alz.org>

- **Alzheimer's Association Professional Resources:** This is a comprehensive hub providing clinicians with the latest evidence-based tools, including:
 - **Clinical Practice Guidelines & Appropriate Use Criteria (AUC):** Guidance on the use of new blood-based biomarkers, amyloid PET, and anti-amyloid treatments like Donanemab.
 - **Assessment & Care Planning Tools:** Resources on utilizing CPT code 99483 for cognitive assessment and developing person-centered care plans.
 - **Educational Materials:** Conversation guides for discussing anti-amyloid treatments and instructional videos for delivering a diagnosis
- **Center to Advance Palliative Care (CAPC) Dementia Care Best Practices:** CAPC offers a toolkit with validated assessment tools, online continuing education courses, and resources focused on **palliative and supportive care** for advanced dementia and caregiver support.
- **Alzheimers.gov** (National Institute on Aging/NIH): This federal government resource offers information for healthcare providers, including details on clinical trials, public health data, and resources for diverse patient populations.
- **AAFP and ACP Guidelines:** The American Academy of Family Physicians (AAFP) often publishes **clinical review articles and guidelines** (like the *Treatment of Alzheimer Disease*) that provide practical, evidence-based recommendations on pharmacological and non-pharmacological interventions for use in primary care.